WORKED SOLUTION

STEP II 1988, Question 10

Pure • Vector Geometry of Paraboloids

STEP II 1988, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q10

The key idea

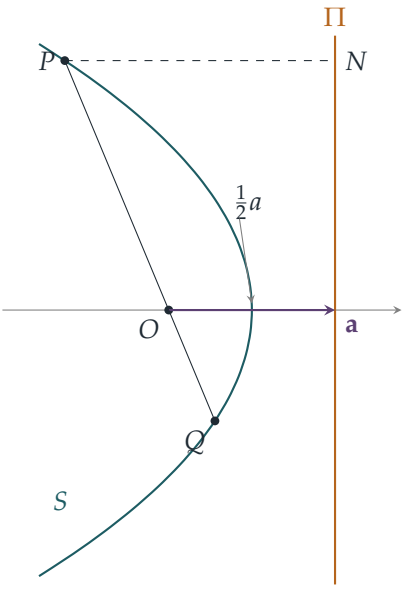
The equation has two pieces with clear geometric meanings: r is the distance from O, and $(a^2 - \mathbf{a}\cdot\mathbf{r})/a$ is the signed distance from the plane $\mathbf{r}\cdot\mathbf{a} = a^2$. So S is a paraboloid of revolution with focus O and that plane as its directrix. For the rest, work along lines through O: write a point on line OP as $\lambda\mathbf{p}$, and a point on the line through O perpendicular to $\mathbf{a}$ and $\mathbf{p}$ as $\mu\,\mathbf{a}\times\mathbf{p}$. Then substitute into the equation of S. The sign of λ matters, because $|\lambda\mathbf{p}| = |\lambda|p$.

Solution

The locus

Let Π be the plane $\mathbf{r}\cdot\mathbf{a} = a^2$. It has unit normal $\hat{\mathbf{a}} = \mathbf{a}/a$, and it lies at distance a from O, on the side towards $\mathbf{a}$. The perpendicular distance from the point $\mathbf{r}$ to Π is

$$d(\mathbf{r}) = |\,a - \hat{\mathbf{a}}\cdot\mathbf{r}\,| = \frac{|a^2 - \mathbf{a}\cdot\mathbf{r}|}{a}.$$



The diagram shows the cross-section of S by a plane containing $\mathbf{a}$. The point P is equally far from O and from its foot N on Π. The surface crosses the axis at $\frac{1}{2}\mathbf{a}$, halfway between O and Π.

Points of S are equidistant. If $\mathbf{a}\cdot\mathbf{r} + ar = a^2$, then $a^2 - \mathbf{a}\cdot\mathbf{r} = ar \geq 0$. So

$$d(\mathbf{r}) = \frac{a^2 - \mathbf{a}\cdot\mathbf{r}}{a} = \frac{ar}{a} = r = |OR|.$$

Equidistant points lie on S. Suppose $r = d(\mathbf{r})$, that is, $ar = |a^2 - \mathbf{a}\cdot\mathbf{r}|$. If $a^2 - \mathbf{a}\cdot\mathbf{r} \geq 0$, this is exactly the equation of S. Otherwise $\mathbf{a}\cdot\mathbf{r} > a^2$, and the condition says $\mathbf{a}\cdot\mathbf{r} = ar + a^2 > ar$. That contradicts $\mathbf{a}\cdot\mathbf{r} \leq ar$ (Cauchy–Schwarz). So no point beyond Π is equidistant, and S is exactly the set of points equidistant from O and Π. This is a paraboloid of revolution about the line through O in the direction of $\mathbf{a}$.

Two consequences are useful later. For P on S,

$$\mathbf{a}\cdot\mathbf{p} = a^2 - ap. \qquad (*)$$

Also $a^2 = \mathbf{a}\cdot\mathbf{p} + ap \leq 2ap$, so $p \geq \frac{1}{2}a$, and hence $2p - a \geq 0$. Equality holds only when $\mathbf{p}$ is a positive multiple of $\mathbf{a}$, that is, when P is the vertex $\frac{1}{2}\mathbf{a}$.

The second intersection Q

Points on line OP are $\lambda\mathbf{p}$, with $|\lambda\mathbf{p}| = |\lambda|p$. Substituting into the equation of S gives

$$\lambda\,\mathbf{a}\cdot\mathbf{p} + a|\lambda|p = a^2.$$

Ask yourself Why can't Q be on the same side of O as P?

If $\lambda > 0$, the left-hand side is $\lambda(\mathbf{a}\cdot\mathbf{p} + ap) = \lambda a^2$, so $\lambda = 1$, which is P itself. (Geometrically, a ray from the focus meets the paraboloid at most once.) So Q has $\lambda < 0$, and then $|\lambda| = -\lambda$. Using $(*)$,

$$\lambda(\mathbf{a}\cdot\mathbf{p} - ap) = a^2 \quad\Longrightarrow\quad \lambda(a^2 - 2ap) = a^2 \quad\Longrightarrow\quad \lambda = -\frac{a}{2p-a}.$$

This is negative, as required, provided $2p > a$, that is, provided P is not the vertex. (At the vertex, the line OP is the axis, and it meets S only once.) Hence

$$\boxed{\mathbf{q} = -\frac{a}{2p-a}\,\mathbf{p}} \qquad \text{and} \qquad |OQ| = \frac{ap}{2p-a}.$$

The points T and T'

The line through O perpendicular to both $\mathbf{a}$ and $\mathbf{p}$ is $\mathbf{t} = \mu\,\mathbf{a}\times\mathbf{p}$. (We need $\mathbf{a}\times\mathbf{p} \neq \mathbf{0}$, so P is not on the axis.) Since $\mathbf{a}\cdot\mathbf{t} = 0$, the equation of S reduces to $at = a^2$, that is, $|\mathbf{t}| = a$. So T and T' lie on the circle where S meets the plane through O parallel to Π, and

$$|\mu|\,|\mathbf{a}\times\mathbf{p}| = a.$$

Ask yourself How do we find $|\mathbf{a}\times\mathbf{p}|$ without knowing the angle between $\mathbf{a}$ and $\mathbf{p}$?

Use the identity $|\mathbf{a}\times\mathbf{p}|^2 = a^2p^2 - (\mathbf{a}\cdot\mathbf{p})^2$, which is $\sin^2\theta = 1 - \cos^2\theta$ in disguise. Then substitute $(*)$:

$$\begin{aligned}|\mathbf{a}\times\mathbf{p}|^2 &= a^2p^2 - (a^2 - ap)^2\\ &= a^2p^2 - a^4 + 2a^3p - a^2p^2\\ &= a^3(2p-a).\end{aligned}$$

Therefore

$$|\mu| = \frac{a}{a^{3/2}\sqrt{2p-a}} = \frac{1}{\sqrt{a(2p-a)}} = \frac{1}{\sqrt{2ap-a^2}},$$

and the position vectors of T and T' are $\pm\dfrac{1}{\sqrt{2ap-a^2}}\,\mathbf{a}\times\mathbf{p}$, as required.

The area of triangle PQT

The points P, O and Q are collinear, with O between P and Q. The vector $\mathbf{t}$ is parallel to $\mathbf{a}\times\mathbf{p}$, so it is perpendicular to $\mathbf{p}$ and hence to the line PQ. So OT is the altitude of triangle PQT from T, and its length is a. The base is

$$|PQ| = p + \frac{ap}{2p-a} = \frac{p(2p-a)+ap}{2p-a} = \frac{2p^2}{2p-a}.$$

Hence

$$\text{area} = \tfrac{1}{2}\cdot|PQ|\cdot|OT| = \tfrac{1}{2}\cdot\frac{2p^2}{2p-a}\cdot a = \frac{ap^2}{2p-a},$$

as required. The same holds with T' in place of T.

Checks

- Take $\mathbf{a} = (2,0,0)$ and $P = (-1.25, 3, 0)$, which is the point in the diagram. Then $\mathbf{a}\cdot\mathbf{p} + ap = -2.5 + 2(3.25) = 4 = a^2$, so P is on S. The formula gives $\mathbf{q} = -\frac{2}{4.5}\mathbf{p} = (0.556, -1.333, 0)$, and $\mathbf{a}\cdot\mathbf{q} + aq = 1.111 + 2.889 = 4$, so Q is on S too.
- Take P at an end of the latus rectum, so $\mathbf{p} \perp \mathbf{a}$ and $p = a$. Then $\mathbf{q} = -\mathbf{p}$, which is right by symmetry. The area is $\frac{1}{2}(2a)(a) = a^2$, and the formula gives $a \cdot a^2/a = a^2$.
- The two focal distances satisfy $\dfrac{1}{OP} + \dfrac{1}{OQ} = \dfrac{1}{p} + \dfrac{2p-a}{ap} = \dfrac{2}{a}$, the classical focal-chord property of a parabola.
- Numerically, we checked Q, T, T' and the area for 20 000 random choices of $\mathbf{a}$ and P. The largest error was about 10^{-11}.

Common slips

- Writing $|\lambda\mathbf{p}| = \lambda p$ for every λ. Then the only solution is $\lambda = 1$, and Q can't be found.
- Forgetting the converse in the first part. A point beyond Π has $|a^2 - \mathbf{a}\cdot\mathbf{r}| = \mathbf{a}\cdot\mathbf{r} - a^2$, and we have to rule out such points.
- Computing the area with a cross product of $\overrightarrow{PQ}$ and $\overrightarrow{PT}$. It works, but it is much longer than noticing that $OT \perp PQ$.

Beyond STEP

This is the focus–directrix definition of a conic, in three dimensions and coordinate-free. With e the eccentricity, the surface $e\,\mathbf{a}\cdot\mathbf{r} + ar = e\,a^2$ (distance from O equal to e times distance from Π) gives the other quadrics of revolution with a focus at O. The relation $1/OP + 1/OQ = 2/a$ is the vector form of the polar equation $\ell/r = 1 + e\cos\theta$, which is central to orbital mechanics. Strang's *Introduction to Linear Algebra* is a good next step for the vector algebra, and Siklos's *Advanced Problems in Mathematics* has more questions in this style.

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